

Problem

Questions:

1. A man possesses five coins, two of which are double-headed, one is double-tailed, and two are normal. He shuts his eyes, picks a coin at random, and tosses it. What is the probability that the lower face of the coin is a head?

He opens his eyes and sees that the coin is showing heads; what is the probability that the lower face is a head?

He shuts his eyes again, and tosses the coin again. What is the probability that the lower face is a head?

He opens his eyes and sees that the coin is showing heads; what is the probability that the lower face is a head?

He discards this coin, picks another at random, and tosses it. What is the probability that it shows heads?

2. A random number N of dice is thrown. Let A_i be the event that $N = i$, and assume that $P(A_i) = 2^{-i}$, $i \geq 1$. The sum of the scores is S . Find the probability that:

- (a) $N = 2$ given $S = 4$;
- (b) $S = 4$ given N is even;
- (c) $N = 2$, given that $S = 4$ and the first die showed 1;
- (d) the largest number shown by any die is r , where S is unknown

3. The event A is said to be repelled by the event B if $P(A | B) < P(A)$, and to be attracted by B if $P(A | B) > P(A)$. Show that if B attracts A , then A attracts B , and B^c repels A . If A attracts B , and B attracts C , does A attract C ?

4. Jane has three children, each of whom is equally likely to be a boy or a girl, independently of the others. Define the following events:

- A : all the children are of the same sex,
- B : there is at most one boy,
- C : the family includes a boy and a girl.

(a) Show that A is independent of B , and that B is independent of C .

(b) Is A independent of C ?

(c) Do these results hold if boys and girls are not equally likely?

(d) Do these results hold if Jane has four children?

5. Three boxes contain balls as follows:

3 red and 2 blue balls

4 red and 3 blue balls

2 red and 5 blue balls

One of the three boxes is selected at random. From the selected box, two balls are drawn successively without replacement.

You are told that at least one of the two balls drawn is red.

Now, before revealing the colours of the two balls, a fair coin is tossed:

If Heads occurs, the first drawn ball is inspected.

If Tails occurs, the second drawn ball is inspected.

The inspected ball is found to be red.

(a) Find the probability that the selected box was B3.

(b) Given that the inspected ball is red, find the probability that both drawn balls were red.

(c) Given that the inspected ball is red, find the probability that the first ball was red and the second ball was blue.

6. Let X have distribution function

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ \frac{1}{2}x & \text{if } 0 \leq x \leq 2, \\ 1 & \text{if } x > 2, \end{cases}$$

and let $Y = X^2$. Find:

(a) $P\left(\frac{1}{2} \leq X \leq \frac{3}{2}\right)$,

(b) $P(1 \leq X < 2)$,

(c) $P(Y \leq X)$,

(d) $P(X \leq 2Y)$,

(e) $P\left(X + Y \leq \frac{3}{4}\right)$,

(f) the distribution function of $Z = \sqrt{X}$.